NVMe/FC Host Configuration for RHEL 8.2 with ONTAP

ONTAP SAN Host

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January 21, 2021

This PDF was generated from https://docs.netapp.com/us-en/ontap-sanhost/nvme_rhel_82.html on March 06, 2021. Always check docs.netapp.com for the latest.

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NVMe/FC Host Configuration for RHEL 8.2 with ONTAP

Supportability

NVMe/FC is supported on ONTAP 9.6 or later for RHEL 8.2. The RHEL 8.2 host runs both NVMe and SCSI traffic through the same fibre channel (FC) initiator adapter ports. See the Hardware Universe for a list of supported FC adapters and controllers. For the most current list of supported configurations and versions, see the NetApp Interoperability Matrix.

Known limitations

For RHEL 8.2, in-kernel NVMe multipath remains disabled by default. Therefore, you need to enable it manually. Steps for doing so are provided in the next section, "Enabling NVMe/FC".

Enabling NVMe/FC

1. Install Red Hat Enterprise Linux 8.2 GA on the server.

   If you are upgrading from RHEL 8.1 to RHEL 8.2 using yum update/upgrade, your /etc/nvme/host* files might be lost. To avoid file loss, do the following:

   a. Backup your /etc/nvme/host* files.

   b. If you have a manually edited udev rule at `/lib/udev/rules.d/71-nvme-iopolicy-netapp-ONTAP.rules`, remove it.

   c. Perform the upgrade

   d. After the upgrade is complete, run the following command:

   ```
   yum remove nvme-cli
   ```

   e. Restore the host files at /etc/nvme/

   ```
   yum install nvmecli
   ```

   f. Copy the original /etc/nvme/host* contents from the backup to the actual host files at /etc/nvme/.

2. Verify that you are running the specified Red Hat Enterprise Linux kernel.

   ```
   # uname -r
   4.18.0-193.el8.x86_64
   ```

   See the NetApp Interoperability Matrix for the most current list of supported versions.

3. Install the nvme-cli package.

```
# rpm -qa|grep nvme-cli
nvme-cli-1.9.5.el8.x86_64
```

4. Enable in-kernel NVMe multipath.

```
# grubby –args=nvme_core.multipath=Y –update-kernel /boot/vmlinuz-
4.18.0-193.el8.x86_64
```

5. On the RHEL 8.2 host, check the host NQN string at /etc/nvme/hostnqn and verify that it matches the host NQN string for the corresponding subsystem on the ONTAP array.

```
# cat /etc/nvme/hostnqn
nqn.2014-08.org.nvmexpress:uuid:9ed5b327-b9fc-4cf5-97b3-1b5d986345d1


::> vserver nvme subsystem host show -vserver vs_fcnvme_141
Vserver      Subsystem        Host           NQN
----------- --------------- ----------- ---------------
  vs_fcnvme_141
    nvme_141_1
        nqn.2014-08.org.nvmexpress:uuid:9ed5b327-b9fc-4cf5-97b3-
1b5d986345d1
```

If the host NQN strings do not match, use the `vserver modify` command to update the host NQN string on the corresponding ONTAP array subsystem to match to host NQM string from /etc/nvme/hostnqn on the host.

6. Reboot the host.
7. Update the `enable_foreign` setting *(optional)*.

If you intend to run both NVMe and SCSI traffic on the same RHEL 8.2 co-existent host, we recommend you use in-kernel NVMe multipath for ONTAP namespaces and dm-multipath for ONTAP LUNs, respectively. You should aslo blacklist the ONTAP namespaces in dm-multipath to prevent dm-multipath from claiming these namespace devices. You do this by adding the `enable_foreign` setting to the /etc/multipath.conf, as shown below.

```
# cat /etc/multipath.conf
defaults {
   enable_foreign NONE
}
```

8. Restart the multipathd daemon by running a `systemctl restart multipathd`.

Validating NVMe/FC

1. Verify the following NVMe/FC settings.

```
# cat /sys/module/nvme_core/parameters/multipath
Y
```

```
# cat /sys/class/nvme-subsystem/nvme-subsys*/model
NetApp ONTAP Controller
NetApp ONTAP Controller
```

```
# cat /sys/class/nvme-subsystem/nvme-subsys*/iopolicy
round-robin
round-robin
```

2. Verify that the namespaces are created.

```
# nvme list
Node SN Model Namespace Usage Format FW Rev
---------------- -------------------- ----------------------
/dev/nvme0n1 80BADBKnB/JvAAAAAAAC NetApp ONTAP Controller 1 53.69 GB /
53.69 GB 4 KiB + 0 B FFFFFFFF
```

3. Verify the status of the ANA paths.

```
# nvme list-subsys/dev/nvme0n1
Nvme-subsysf0 - NQN=nqn.1992-
08.com.netapp:sn.341541339b9511e8a9b500a098c80f09:subsystem.rhel_141_nvm
e_ss_10_0
\
+- nvme0 fc traddr=nn-0x202c00a098c80f09:pn-0x202d00a098c80f09
host_traddr=nn-0x20000090fae0ec61:pn-0x10000090fae0ec61 live optimized
+- nvme1 fc traddr=nn-0x207300a098dfdd91:pn-0x207600a098dfdd91
host_traddr=nn-0x200000109b1c1204:pn-0x100000109b1c1204 live
inaccessible
+- nvme2 fc traddr=nn-0x207300a098dfdd91:pn-0x207500a098dfdd91
host_traddr=nn-0x200000109b1c1205:pn-0x100000109b1c1205 live optimized
+- nvme3 fc traddr=nn-0x207300a098dfdd91:pn-0x207700a098dfdd91 host
traddr=nn-0x200000109b1c1205:pn-0x100000109b1c1205 live inaccessible
```

4. Verify the NetApp plug-in for ONTAP devices.

```
# nvme netapp ontapdevices -o column
Device   Vserver  Namespace Path             NSID   UUID   Size
-------  -------- -------------------------  ------ ----- -----
/dev/nvme0n1   vs_nvme_10       /vol/rhel_141_vol_10_0/rhel_141_ns_10_0
1        55baf453-f629-4a18-9364-b6aee3f50dad   53.69GB

# nvme netapp ontapdevices -o json
{
   "ONTAPdevices" : [
   {
        Device" : "/dev/nvme0n1",
        "Vserver" : "vs_nvme_10",
        "Namespace_Path" : "/vol/rhel_141_vol_10_0/rhel_141_ns_10_0",
         "NSID" : 1,
         "UUID" : "55baf453-f629-4a18-9364-b6aee3f50dad",
         "Size" : "53.69GB",
         "LBA_Data_Size" : 4096,
         "Namespace_Size" : 13107200
    }
]
```

Configuring the Broadcom FC Adapter

For the most current list of supported adapters see the see the NetApp Interoperability Matrix.

1. Verify that you are using the supported adapter.

```
# cat /sys/class/scsi_host/host*/modelname
LPe32002-M2
LPe32002-M2
```

```
# cat /sys/class/scsi_host/host*/modeldesc
Emulex LightPulse LPe32002-M2 2-Port 32Gb Fibre Channel Adapter
Emulex LightPulse LPe32002-M2 2-Port 32Gb Fibre Channel Adapter
```

2. Verify that you are using the recommended Broadcom lpfc firmware as well as the inbox driver.

```
# cat /sys/class/scsi_host/host*/fwrev
12.6.182.8, sli-4:2:c
12.6.182.8, sli-4:2:c
```

```
# cat /sys/module/lpfc/version
0:12.6.0.2
```

3. Verify that `lpfc_enable_fc4_type` is set to "**3**".

```
# cat /sys/module/lpfc/parameters/lpfc_enable_fc4_type
3
```

4. Verify that the initiator ports are up and running and are able to see the target LIFs.

```
# cat /sys/class/fc_host/host*/port_name
0x100000109b1c1204
0x100000109b1c1205
```

```
# cat /sys/class/fc_host/host*/port_state
Online
Online
```

```
# cat /sys/class/scsi_host/host*/nvme_info
NVME Initiator Enabled
XRI Dist lpfc0 Total 6144 IO 5894 ELS 250
NVME LPORT lpfc0 WWPN x100000109b1c1204 WWNN x200000109b1c1204 DID
x011d00 ONLINE
NVME RPORT WWPN x203800a098dfdd91 WWNN x203700a098dfdd91 DID x010c07
TARGET DISCSRVC ONLINE
NVME RPORT WWPN x203900a098dfdd91 WWNN x203700a098dfdd91 DID x011507
TARGET DISCSRVC ONLINE
NVME Statistics
LS: Xmt 0000000f78 Cmpl 0000000f78 Abort 00000000
LS XMIT: Err 00000000 CMPL: xb 00000000 Err 00000000
Total FCP Cmpl 000000002fe29bba Issue 000000002fe29bc4 OutIO
000000000000000a
abort 00001bc7 noxri 00000000 nondlp 00000000 qdepth 00000000 wqerr
00000000 err 00000000
FCP CMPL: xb 00001e15 Err 0000d906
NVME Initiator Enabled
XRI Dist lpfc1 Total 6144 IO 5894 ELS 250
NVME LPORT lpfc1 WWPN x100000109b1c1205 WWNN x200000109b1c1205 DID
x011900 ONLINE
NVME RPORT WWPN x203d00a098dfdd91 WWNN x203700a098dfdd91 DID x010007
TARGET DISCSRVC ONLINE
NVME RPORT WWPN x203a00a098dfdd91 WWNN x203700a098dfdd91 DID x012a07
TARGET DISCSRVC ONLINE
NVME Statistics
LS: Xmt 0000000fa8 Cmpl 0000000fa8 Abort 00000000
LS XMIT: Err 00000000 CMPL: xb 00000000 Err 00000000
Total FCP Cmpl 000000002e14f170 Issue 000000002e14f17a OutIO
000000000000000a
abort 000016bb noxri 00000000 nondlp 00000000 qdepth 00000000 wqerr
00000000 err 00000000
FCP CMPL: xb 00001f50 Err 0000d9f8
```

5. Enable 1 MB I/O size *(optional)*.

   The `lpfc_sg_seg_cnt` parameter needs to be set to 256 for the lpfc driver to issue I/O requests up to 1 MB in size.

```
# cat /etc/modprobe.d/lpfc.conf
options lpfc lpfc_sg_seg_cnt=256
```

6. Run a `dracut -f` command and then reboot the host.
7. After the host boots up, verify that lpfc_sg_seg_cnt is set to 256.

```
# cat /sys/module/lpfc/parameters/lpfc_sg_seg_cnt
256
```

LPFC Verbose Logging

1. You can set the lpfc_log_verbose driver setting to any of the following values to log NVMe/FC events.

```
#define LOG_NVME 0x00100000 /* NVME general events. */
#define LOG_NVME_DISC 0x00200000 /* NVME Discovery/Connect events. */
#define LOG_NVME_ABTS 0x00400000 /* NVME ABTS events. */
#define LOG_NVME_IOERR 0x00800000 /* NVME IO Error events. */
```

2. After setting any of these values, run `dracut-f` and reboot host.
3. After rebooting, verify the settings.

```
# cat /etc/modprobe.d/lpfc.conf
lpfc_enable_fc4_type=3 lpfc_log_verbose=0xf00083

# cat /sys/module/lpfc/parameters/lpfc_log_verbose
15728771
```

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